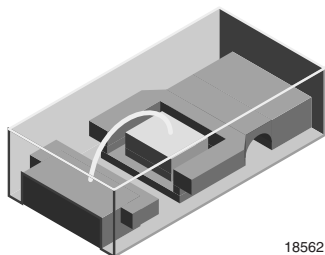


## Low Current 0603 SMD LED



### DESCRIPTION

The new 0603 LED series have been designed in the smallest SMD package. This innovative 0603 LED technology opens the way to

- Smaller products of higher performance
- More design in flexibility
- Enhanced applications

The 0603 LED is an obvious solution for small-scale, high power products that are expected to work reliability in an arduous environment.

### PRODUCT GROUP AND PACKAGE DATA

- Product group: LED
- Package: SMD 0603
- Product series: low current
- Angle of half intensity:  $\pm 80^\circ$

### FEATURES

- Smallest SMD package 0603 with exceptional brightness 1.6 mm x 0.8 mm x 0.6 mm (L x W x H)
- High reliability lead frame based
- Temperature range  $-40^\circ\text{C}$  to  $+100^\circ\text{C}$
- Footprint compatible to 0603 chipled
- Wavelength 633 nm (red), 606 nm (orange), 587 nm (yellow)
- AlInGaP technology
- Compatible to IR reflow soldering
- Viewing angle: Extremely wide  $160^\circ$
- Grouping parameter: Luminous intensity, wavelength
- Available in 8 mm tape
- Preconditioning according to JEDEC® level 2
- AEC-Q101 qualified
- Material categorization: for definitions of compliance please see [www.vishay.com/doc?99912](http://www.vishay.com/doc?99912)



### APPLICATIONS

- Backlight keypads
- Navigation systems
- Cellular phone displays
- Displays for industrial control systems
- Automotive features
- Miniaturized color effects
- Traffic displays

### PARTS TABLE

| PART          | COLOR       | LUMINOUS INTENSITY (mcd) |      |      | at $I_F$ (mA) | WAVELENGTH (nm) |      |      | at $I_F$ (mA) | FORWARD VOLTAGE (V) |      |      | at $I_F$ (mA) | TECHNOLOGY |
|---------------|-------------|--------------------------|------|------|---------------|-----------------|------|------|---------------|---------------------|------|------|---------------|------------|
|               |             | MIN.                     | TYP. | MAX. |               | MIN.            | TYP. | MAX. |               | MIN.                | TYP. | MAX. |               |            |
| TLMS1000-GS08 | Red         | 1.8                      | 4    | -    | 2             | 624             | 628  | 636  | 2             | -                   | 1.8  | 2.6  | 2             | AlInGaP    |
| TLMO1000-GS08 | Soft orange | 3.55                     | 7.5  | -    | 2             | 600             | 605  | 609  | 2             | -                   | 1.8  | 2.6  | 2             | AlInGaP    |
| TLMY1000-GS08 | Yellow      | 3.55                     | 7.5  | -    | 2             | 580             | 588  | 595  | 2             | -                   | 1.8  | 2.6  | 2             | AlInGaP    |

### ABSOLUTE MAXIMUM RATINGS ( $T_{amb} = 25^\circ\text{C}$ , unless otherwise specified)

#### TLMS1000, TLMO1000, TLMY1000

| PARAMETER                           | TEST CONDITION                                     | SYMBOL     | VALUE       | UNIT             |
|-------------------------------------|----------------------------------------------------|------------|-------------|------------------|
| Reverse voltage <sup>(1)</sup>      |                                                    | $V_R$      | 12          | V                |
| DC Forward current                  | $T_{amb} \leq 95^\circ\text{C}$                    | $I_F$      | 15          | mA               |
| Surge forward current               | $t_p \leq 10 \mu\text{s}$                          | $I_{FSM}$  | 0.1         | A                |
| Power dissipation                   |                                                    | $P_V$      | 40          | mW               |
| Junction temperature                |                                                    | $T_j$      | 120         | $^\circ\text{C}$ |
| Operating temperature range         |                                                    | $T_{amb}$  | -40 to +100 | $^\circ\text{C}$ |
| Storage temperature range           |                                                    | $T_{stg}$  | -40 to +100 | $^\circ\text{C}$ |
| Soldering temperature               | Acc. Vishay spec                                   | $T_{sd}$   | 260         | $^\circ\text{C}$ |
| Thermal resistance junction/ambient | Mounted on PC board (pad size $> 5 \text{ mm}^2$ ) | $R_{thJA}$ | 500         | K/W              |

#### Note

<sup>(1)</sup> Driving the LED in reverse direction is suitable for short term application

**OPTICAL AND ELECTRICAL CHARACTERISTICS** ( $T_{amb} = 25\text{ }^{\circ}\text{C}$ , unless otherwise specified)  
**TLMS1000, RED**

| PARAMETER               | TEST CONDITION                          | SYMBOL      | MIN. | TYP.     | MAX. | UNIT       |
|-------------------------|-----------------------------------------|-------------|------|----------|------|------------|
| Luminous intensity      | $I_F = 2\text{ mA}$                     | $I_V$       | 1.8  | 4        | -    | mcd        |
| Dominant wavelength     | $I_F = 2\text{ mA}$                     | $\lambda_d$ | 624  | 628      | 636  | nm         |
| Peak wavelength         | $I_F = 2\text{ mA}$                     | $\lambda_p$ | -    | 640      | -    | nm         |
| Angle of half intensity | $I_F = 2\text{ mA}$                     | $\varphi$   | -    | $\pm 80$ | -    | $^{\circ}$ |
| Forward voltage         | $I_F = 2\text{ mA}$                     | $V_F$       | -    | 1.8      | 2.6  | V          |
| Reverse voltage         | $I_R = 10\text{ }\mu\text{A}$           | $V_R$       | 6    | -        | -    | V          |
| Junction capacitance    | $V_R = 0\text{ V}$ , $f = 1\text{ MHz}$ | $C_j$       | -    | 15       | -    | pF         |

**OPTICAL AND ELECTRICAL CHARACTERISTICS** ( $T_{amb} = 25\text{ }^{\circ}\text{C}$ , unless otherwise specified)  
**TLMO1000, SOFT ORANGE**

| PARAMETER               | TEST CONDITION                          | SYMBOL      | MIN. | TYP.     | MAX. | UNIT       |
|-------------------------|-----------------------------------------|-------------|------|----------|------|------------|
| Luminous intensity      | $I_F = 2\text{ mA}$                     | $I_V$       | 3.55 | 7.5      | -    | mcd        |
| Dominant wavelength     | $I_F = 2\text{ mA}$                     | $\lambda_d$ | 600  | 605      | 609  | nm         |
| Peak wavelength         | $I_F = 2\text{ mA}$                     | $\lambda_p$ | -    | 610      | -    | nm         |
| Angle of half intensity | $I_F = 2\text{ mA}$                     | $\varphi$   | -    | $\pm 80$ | -    | $^{\circ}$ |
| Forward voltage         | $I_F = 2\text{ mA}$                     | $V_F$       | -    | 1.8      | 2.6  | V          |
| Reverse voltage         | $I_R = 10\text{ }\mu\text{A}$           | $V_R$       | 6    | -        | -    | V          |
| Junction capacitance    | $V_R = 0\text{ V}$ , $f = 1\text{ MHz}$ | $C_j$       | -    | 15       | -    | pF         |

**OPTICAL AND ELECTRICAL CHARACTERISTICS** ( $T_{amb} = 25\text{ }^{\circ}\text{C}$ , unless otherwise specified)  
**TLMY1000, YELLOW**

| PARAMETER               | TEST CONDITION                          | SYMBOL      | MIN. | TYP.     | MAX. | UNIT       |
|-------------------------|-----------------------------------------|-------------|------|----------|------|------------|
| Luminous intensity      | $I_F = 2\text{ mA}$                     | $I_V$       | 3.55 | 7.5      | -    | mcd        |
| Dominant wavelength     | $I_F = 2\text{ mA}$                     | $\lambda_d$ | 580  | 588      | 595  | nm         |
| Peak wavelength         | $I_F = 2\text{ mA}$                     | $\lambda_p$ | -    | 591      | -    | nm         |
| Angle of half intensity | $I_F = 2\text{ mA}$                     | $\varphi$   | -    | $\pm 80$ | -    | $^{\circ}$ |
| Forward voltage         | $I_F = 2\text{ mA}$                     | $V_F$       | -    | 1.8      | 2.6  | V          |
| Reverse voltage         | $I_R = 10\text{ }\mu\text{A}$           | $V_R$       | 6    | -        | -    | V          |
| Junction capacitance    | $V_R = 0\text{ V}$ , $f = 1\text{ MHz}$ | $C_j$       | -    | 15       | -    | pF         |

**COLOR CLASSIFICATION**

| GROUP | DOMINANT WAVELENGTH (nm) |      |        |      |
|-------|--------------------------|------|--------|------|
|       | YELLOW                   |      | ORANGE |      |
|       | MIN.                     | MAX. | MIN.   | MAX. |
| 2     | 580                      | 583  | 600    | 603  |
| 3     | 583                      | 586  | 602    | 605  |
| 4     | 586                      | 589  | 604    | 607  |
| 5     | 589                      | 592  | 606    | 609  |
| 6     | 592                      | 595  |        |      |

**Note**

- Wavelengths are tested at a current pulse duration of 25 ms and an accuracy of  $\pm 1\text{ nm}$

**LUMINOUS INTENSITY CLASSIFICATION**

| GROUP | LUMINOUS INTENSITY (mcd) |       |
|-------|--------------------------|-------|
|       | MIN.                     | MAX.  |
| G1    | 1.80                     | 2.24  |
| G2    | 2.24                     | 2.80  |
| H1    | 2.80                     | 3.55  |
| H2    | 3.55                     | 4.50  |
| J1    | 4.50                     | 5.60  |
| J2    | 5.60                     | 7.10  |
| K1    | 7.10                     | 9.00  |
| K2    | 9.00                     | 11.20 |
| L1    | 11.20                    | 14.00 |
| L2    | 14.00                    | 18.00 |

**Note**

- Luminous intensity is tested at a current pulse duration of 25 ms and an accuracy of  $\pm 11\%$ .  
The above type numbers represent the order groups which include only a few brightness groups. Only one group will be shipped on each reel (there will be no mixing of two groups on each reel).  
In order to ensure availability, single brightness groups will not be orderable.  
In a similar manner for colors where wavelength groups are measured and binned, single wavelength groups will be shipped in any one reel.  
In order to ensure availability, single wavelength groups will not be orderable

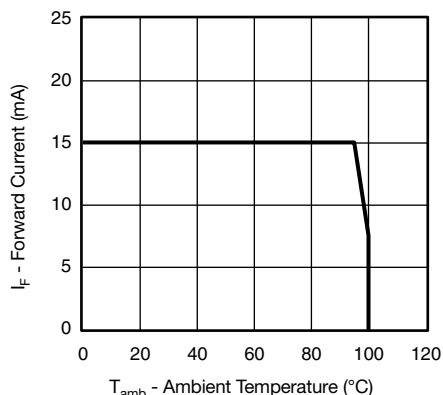
**TYPICAL CHARACTERISTICS** ( $T_{amb} = 25^\circ\text{C}$ , unless otherwise specified)

Fig. 1 - Forward Current vs. Ambient Temperature

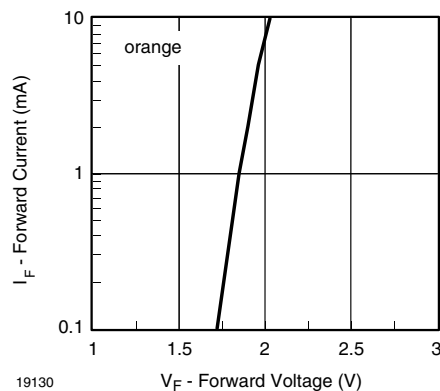


Fig. 3 - Forward Current vs. Forward Voltage

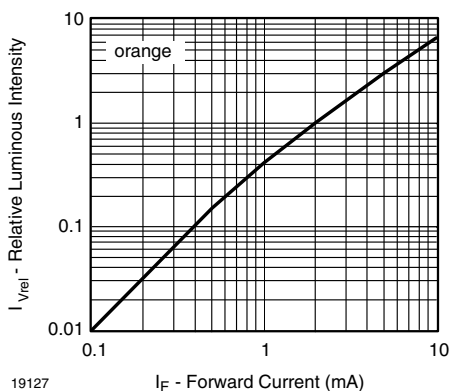


Fig. 2 - Relative Luminous Intensity vs. Forward Current

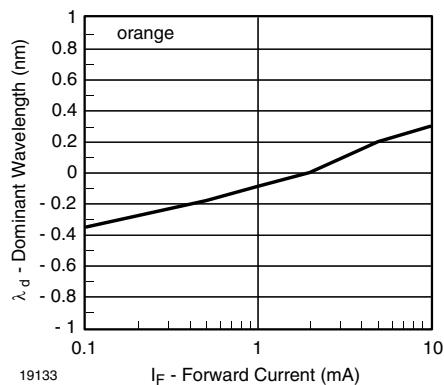


Fig. 4 - Dominant Wavelength vs. Forward Current

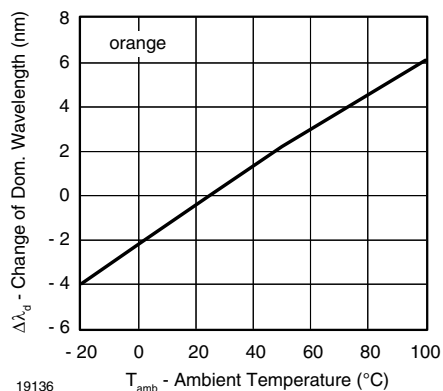


Fig. 5 - Change of Dominant Wavelength vs. Ambient Temperature

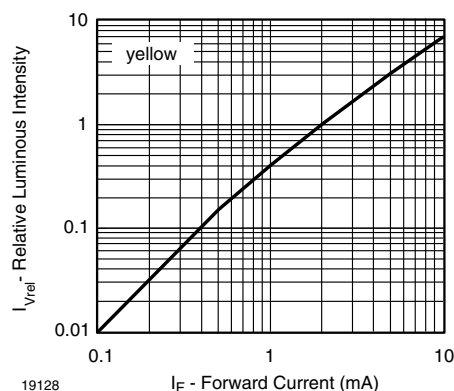


Fig. 8 - Relative Luminous Intensity vs. Forward Current

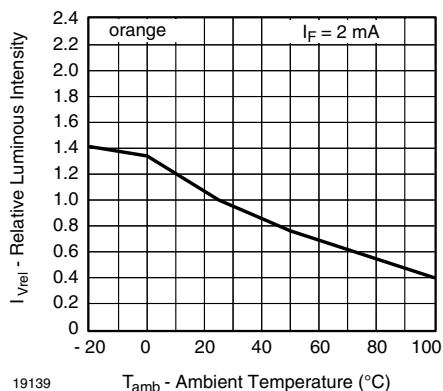


Fig. 6 - Relative Luminous Intensity vs. Ambient Temperature

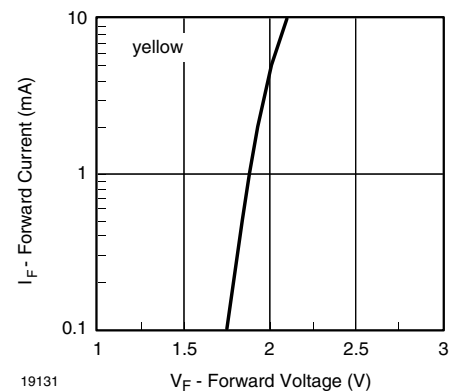


Fig. 9 - Forward Current vs. Forward Voltage

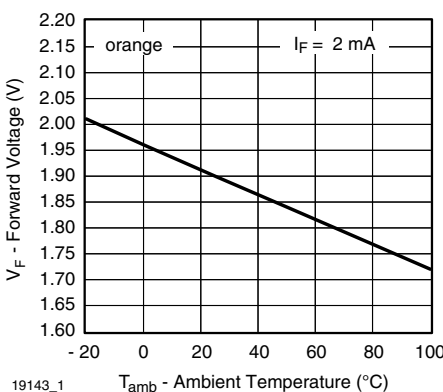


Fig. 7 - Forward Voltage vs. Ambient Temperature

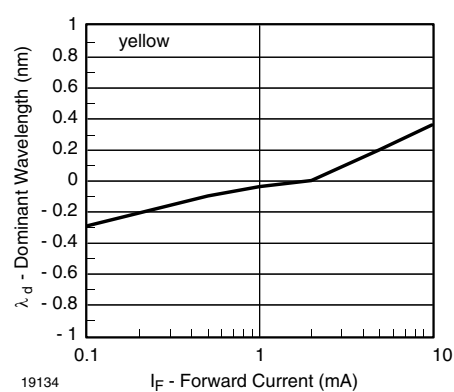


Fig. 10 - Dominant Wavelength vs. Forward Current

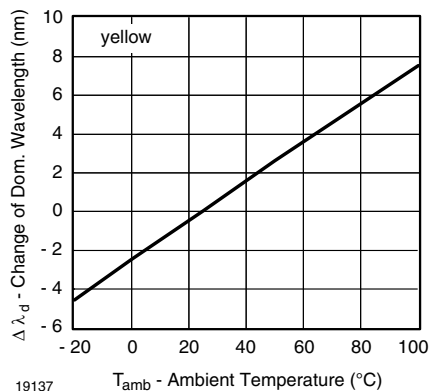


Fig. 11 - Change of Dominant Wavelength vs. Ambient Temperature

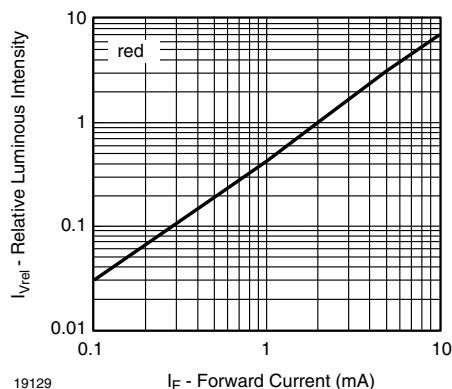


Fig. 14 - Relative Luminous Intensity vs. Forward Current

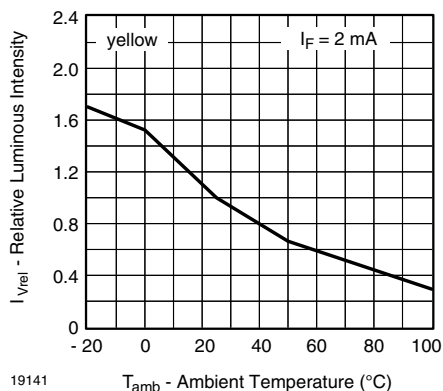


Fig. 12 - Relative Luminous Intensity vs. Ambient Temperature

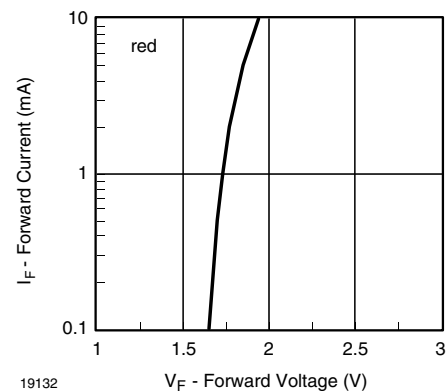


Fig. 15 - Forward Current vs. Forward Voltage

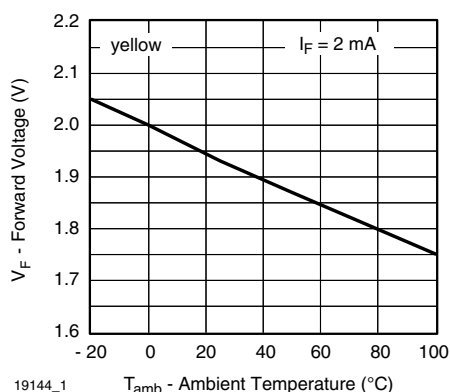


Fig. 13 - Forward Voltage vs. Ambient Temperature

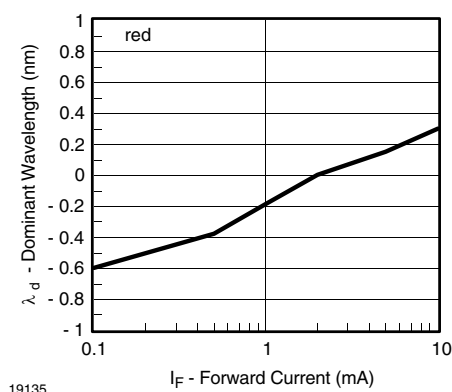


Fig. 16 - Dominant Wavelength vs. Forward Current

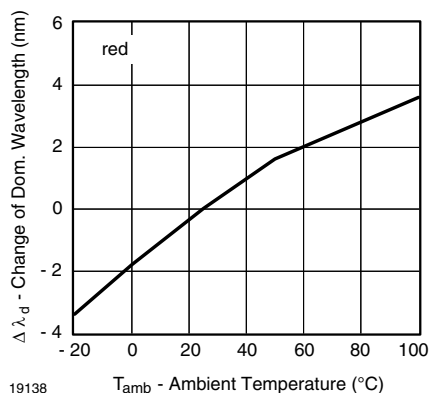


Fig. 17 - Change of Dominant Wavelength vs. Ambient Temperature

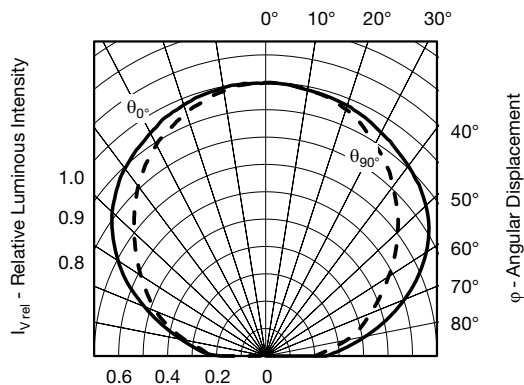


Fig. 20 - Relative Luminous Intensity vs. Angular Displacement

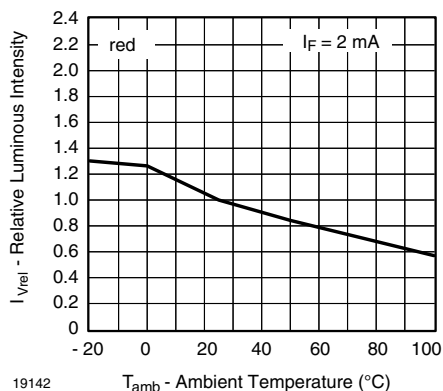


Fig. 18 - Relative Luminous Intensity vs. Ambient Temperature

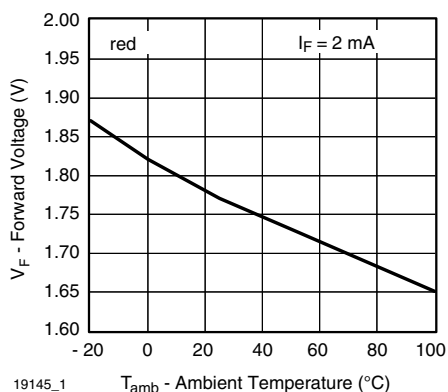
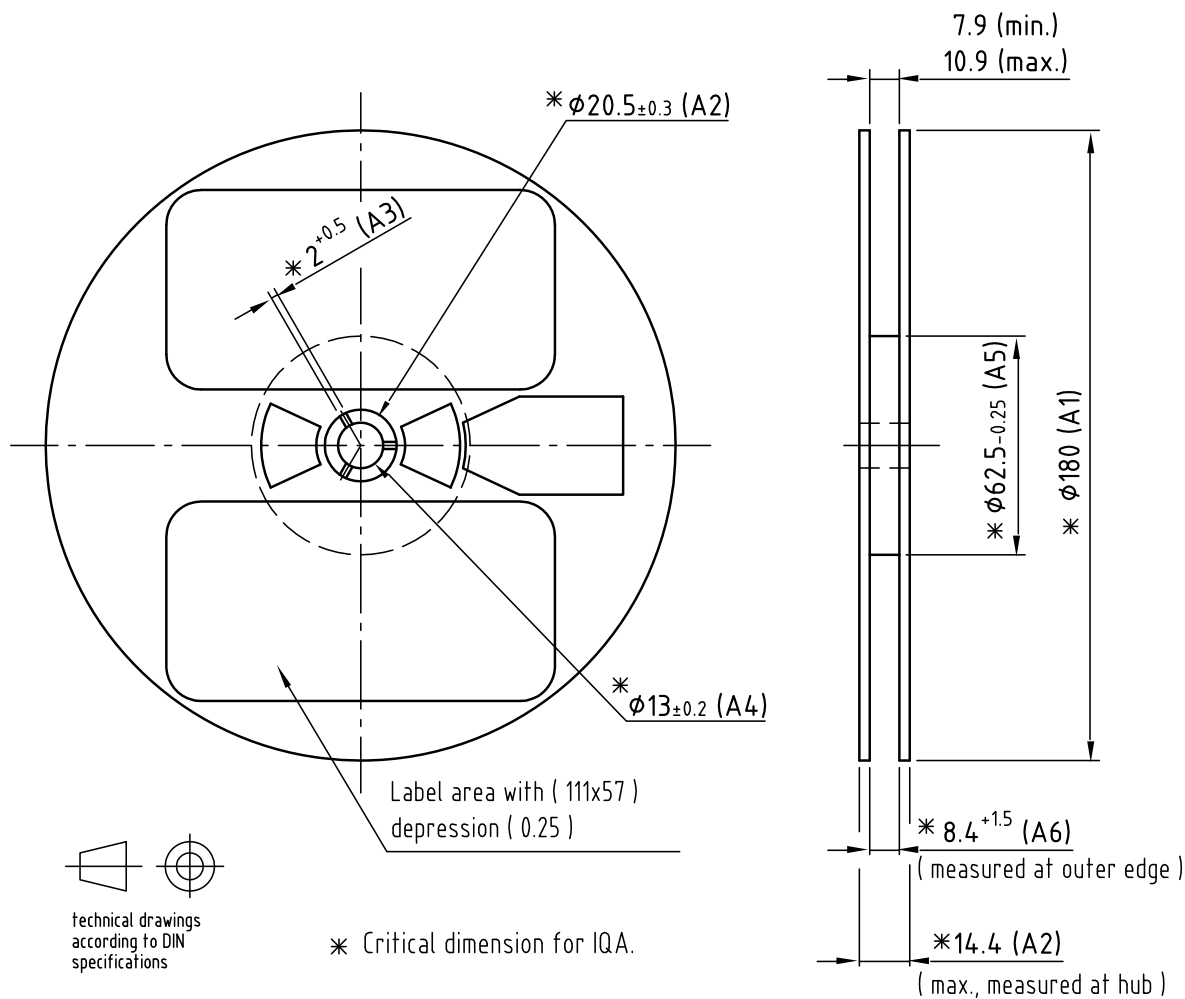


Fig. 19 - Forward Voltage vs. Ambient Temperature



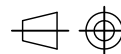
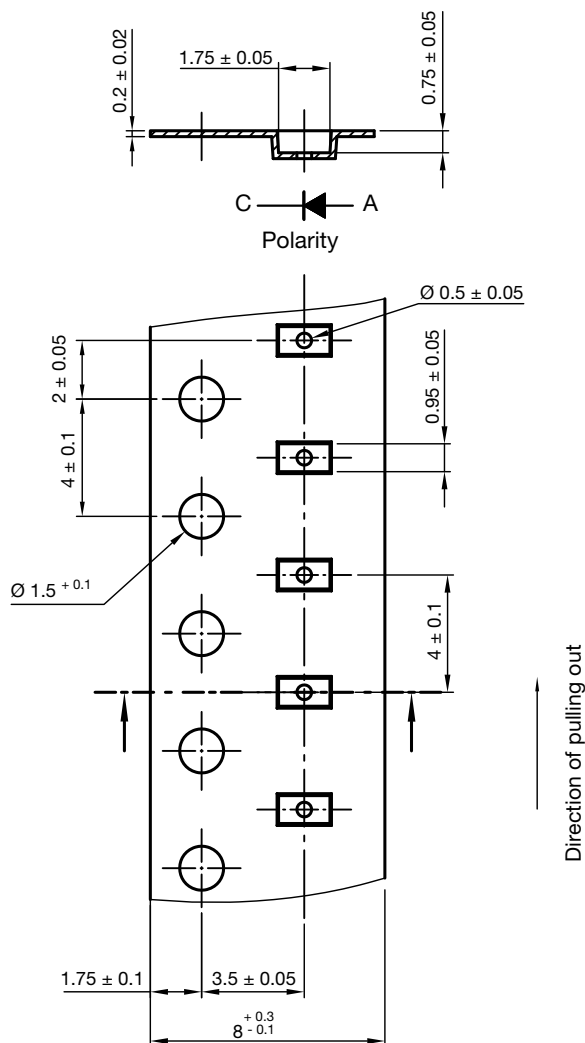
**REEL DIMENSIONS** in millimeters



Drawing-No.: 9.800-5086.01-4  
Issue: 1; 29.04.04  
19043

Not indicated tolerances  $\pm 0.05$   
Material: black static dissipative

### TAPE DIMENSIONS in millimeters

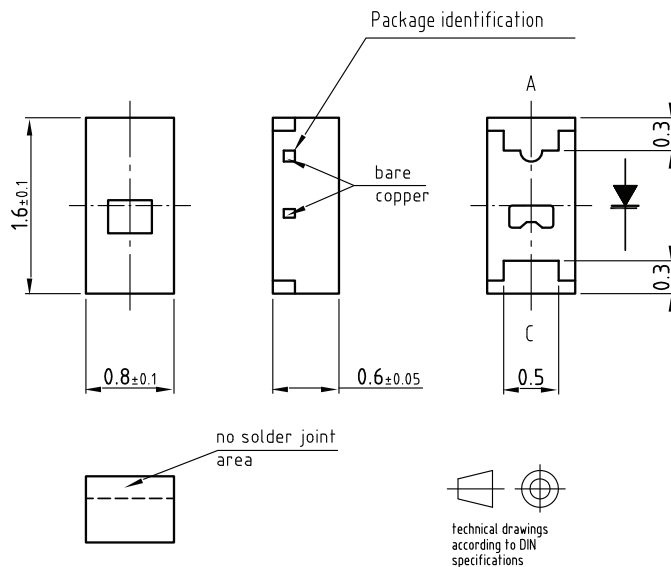


Technical drawings  
according to DIN  
specifications

Not indicated tolerances  $\pm 0.05$   
Material: Conductive black PC

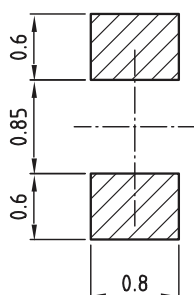
Drawing-No.: 9.700-5290.01-4  
Issue: 3; 24.09.13



**PACKAGE DIMENSIONS** in millimeters


Not indicated tolerances  $\pm 0.1$

Recommended solder pad



Drawing-No.: 6.541-5056.01-4

Issue: 2; 04.05.05

19426

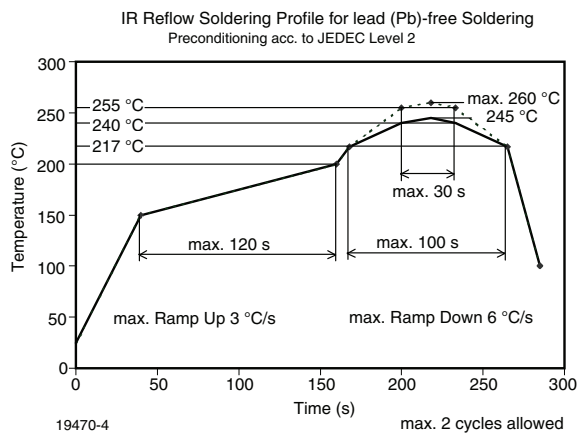
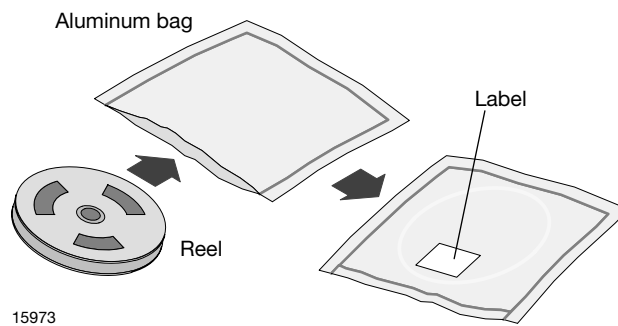
**SOLDERING PROFILE**


Fig. 21 - Vishay Lead (Pb)-free Reflow Soldering Profile (acc. to J-STD-020C)

**DRY PACKING**

The reel is packed in an anti-humidity bag to protect the devices from absorbing moisture during transportation and storage.





## FINAL PACKING

The sealed reel is packed into a cardboard box. A secondary cardboard box is used for shipping purposes.

## RECOMMENDED METHOD OF STORAGE

Dry box storage is recommended as soon as the aluminum bag has been opened to prevent moisture absorption. The following conditions should be observed, if dry boxes are not available:

- Storage temperature 10 °C to 30 °C
- Storage humidity  $\leq 60\%$  RH max.

After more than 1 year under these conditions moisture content will be too high for reflow soldering.

In case of moisture absorption, the devices will recover to the former condition by drying under the following condition:

192 h at 40 °C + 5 °C / - 0 °C and  $< 5\%$  RH (dry air/nitrogen)  
or

96 h at 60 °C + 5 °C and  $< 5\%$  RH for all device containers  
or

24 h at 100 °C + 5 °C not suitable for reel or tubes.

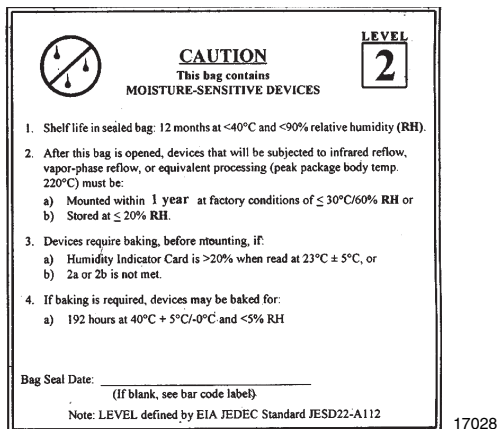
An EIA JEDEC standard JESD22-A112 level 2 label is included on all dry bags.

## ESD PRECAUTION

Proper storage and handling procedures should be followed to prevent ESD damage to the devices especially when they are removed from the antistatic shielding bag. Electro-static sensitive devices warning labels are on the packaging.

## VISHAY SEMICONDUCTORS STANDARD BAR CODE LABEL

The Vishay Semiconductors standard bar code labels are printed at final packing areas. The labels are on each packing unit and contain Vishay Semiconductors specific data.



Example of JESD22-A112 level 2 label



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